

Crash course: how to make your research Bayesian

Syllabus

(1) Bayesian thinking and building intuition 14 Sep - Phys 02.114

- Probability and uncertainty
- Bayes' theorem as a way of updating beliefs
- Prior, likelihood, posterior, evidence
- Discrete examples

(2) Building simple inference models 21 Sep - Phys 02.114

- Continuous probabilities and probability density functions
- Calculating and interpreting 1D posteriors
- Prior dependence and sensitivity to assumptions

(3) More dimensions, comparing models 28 Sep - Phys 02.114

- Joint posteriors in multiple dimensions
- Marginalization over nuisance parameters
- Summarizing posteriors: credible intervals, contour and corner plots
- Comparing competing models

(4) Bayesian renaissance: numerical methods 5 Oct - Phys _____.102

- Why direct posterior evaluation fails in many dimensions
- Likelihoods with no closed form, where every evaluation runs a simulation
- MCMC
- Nested sampling
- A realistic inference example